Cyber Security Project

Phishing website detection system

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Course : BSc. Computer sci.

Title

Phishing Website Detection System

Project Name

Website Security Verification Tool

Objective

The objective of this project is to detect potentially phishing websites by analyzing URL characteristics and identifying suspicious patterns. The system helps users avoid fraudulent websites that attempt to steal sensitive information such as usernames, passwords, and banking details.

Problem Statement

Phishing attacks are among the most common cyber threats. Attackers create fake websites that closely resemble legitimate websites to trick users into revealing confidential information. Many users find it difficult to distinguish between genuine and phishing websites. Therefore, an automated system is needed to identify suspicious websites and warn users before they access them.

Research Findings

- Phishing websites often use unusual URLs, excessive special characters, or shortened links.
- Many phishing URLs contain IP addresses instead of domain names.
- Attackers frequently use misspelled versions of popular website names.
- Machine learning and URL-based analysis are commonly used techniques for phishing detection.
- Early detection of phishing websites can significantly reduce cybercrime and data theft.

Methodology

1. Study common characteristics of phishing websites.
2. Collect examples of legitimate and phishing URLs.
3. Identify URL-based indicators of phishing attacks.
4. Develop a detection system that analyzes URLs.
5. Classify URLs as Safe or Suspicious.
6. Test the system with multiple website links.

Tools Used

- Python 3
- Regular Expressions (re)
- URL Parsing Libraries
- VS Code / PyCharm
- Dataset of Legitimate and Phishing URLs (for testing)

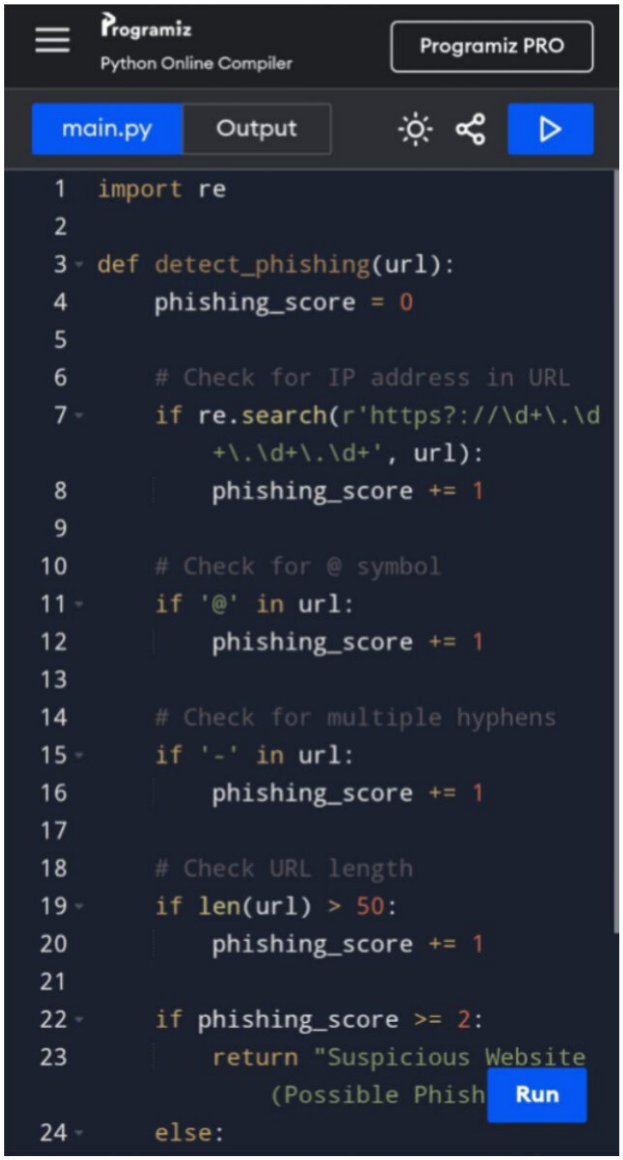
Implementation

Python Code

```
import re

def detect_phishing(url):
    phishing_score = 0

    # Check for IP address in URL
    if re.search(r'https?://\d+\.\d+\.\d+\.\d+', url):
        phishing_score += 1

    # Check for @ symbol
    if '@' in url:
        phishing_score += 1

    # Check for multiple hyphens
    if '-' in url:
        phishing_score += 1

    # Check URL length
    if len(url) > 50:
        phishing_score += 1

    if phishing_score >= 2:
        return "Suspicious Website (Possible Phishing)"
    else:
        return "Safe Website"

url = input("Enter Website URL: ")
result = detect_phishing(url)

print("\nAnalysis Result:")
print(result)
```

Screenshots

Python code

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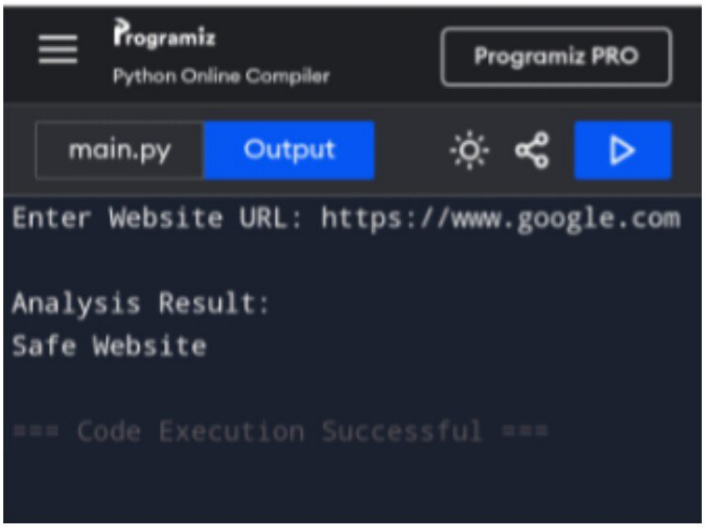
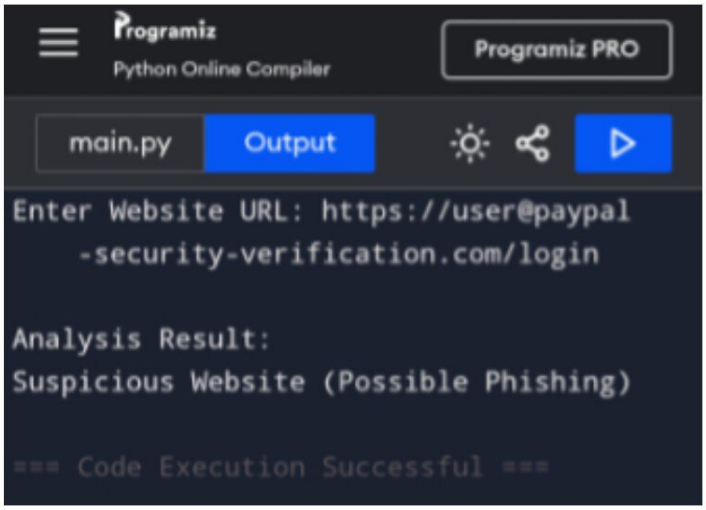


Screenshot
Output



Results

- Successfully analyzed website URLs for phishing indicators.
- Identified suspicious URLs based on predefined security rules.
- Provided instant feedback to users regarding website safety.
- Demonstrated practical application of cybersecurity concepts in phishing prevention.

Challenges

- Some legitimate websites may contain characteristics similar to phishing URLs.
- Advanced phishing websites can bypass simple detection rules.

- Maintaining high detection accuracy requires continuous updates and testing.

Future Scope

- Integrate Machine Learning algorithms for improved detection accuracy.
- Develop a browser extension for real-time phishing protection.
- Add domain reputation and SSL certificate verification.
- Connect with online threat intelligence databases.
- Create a graphical user interface for better usability.

Conclusion

The Phishing Website Detection System is an effective cybersecurity project that helps users identify potentially fraudulent websites. By analyzing URL patterns and suspicious indicators, the system provides an additional layer of protection against phishing attacks. The project demonstrates how cybersecurity techniques can be applied to enhance online safety and protect sensitive information from cybercriminals.